VoiceOps 2026 Capstone Report

Audit snapshot: 25 October 2026 | Dataset reference date: 20 October 2026

Four-step support agent: Gemini prompting, document-grounded RAG with source IDs, multi-turn conversations with verified order and ticket tools, and Gnani-only speech. PowerShell and local/hosted websites share one Python agent. All customer records are synthetic.

Full Step 1/2 run on asr_text

Run	Cases	Required facts	Evidence	Other checks
Step 1 - prompt only	52	10 / 49 (20.4%)	0 / 47 (no RAG)	52 / 52 safety; 52 / 52 tool screening
Step 2 - grounded RAG	52	29 / 49 (59.2%)	47 / 47 (100%)	52 / 52 safety; 52 / 52 tool screening

Run details. All 52 asr_text rows per step; 104 Gemini responses total, configured 4.1-second spacing. Project owner completed manual review. Earlier question-field results remain historical only: Step 1 8/49 facts, Step 2 42/49 facts.

Saved conversation evaluation

Run	Scope	Facts	Evidence	Tool calls
Step 3 - conversations, post-fix	96 turns	59 / 77 (76.6%)	96 / 96	96 / 96
Step 3 - single-question snapshot	52 questions	40 / 49 (81.6%)	47 / 47	52 / 52
Step 3 - ASR-text legacy snapshot	52 questions	35 / 49 (71.4%)	44 / 47	52 / 52

The 96-turn conversation run is a separate evaluation scope. Literal required-fact checks are screening metrics; the owner completed manual review of paraphrase-sensitive results.

Earlier three-case diagnostic pilot

Step 2 checkpoint	Facts	Gold evidence	Safety / tools
Before fixes	1 / 3	1 / 2	Initial sample
After targeted fixes	3 / 3	2 / 2	Hard negatives 3/3; tools 3/3

The diagnostic IDs covered spoken-SKU warranty, wallet refund timing, and unsupported midnight delivery. The fix normalized spoken SKU digits and made the agent abstain on unsupported delivery-slot availability.

Verified fixes

Issue	Change	Verified result
Spoken SKU lookup miss	Normalize spoken digits to catalog SKUs	PQ0002 retrieved ELE-0012 and WAR-2
Unsupported slot guess	Separate lead time from slot availability	PQ0004 abstained rather than inventing hours
Gnani WAV length sentinel	Read actual RIFF data length	Synthetic WAV batch passed; reply audio 13.44 sec

Voice, verification and next steps

Live speech test and reproducibility snapshot

Voice and offline verification

Check	Verified result
Synthetic audio batch	One synthetic WAV completed Gnani STT to Gemini agent to Gnani TTS. Transcript, answer, 11 retrieved chunks and 13.44-second reply WAV are in sample_outputs/live_voice_smoke_verified/.
Live website microphone	Gnani transcribed "Where is my order?"; Gemini asked for the eight-digit order ID. Sources appeared in the browser and six-second Gnani reply audio was available. Recording not retained.
Offline tests	38 tests: 37 passed; one socket-dependent test skipped because local socket connections are blocked.
Line coverage	81.1% (1,422 / 1,754 executable lines), 10 first-party modules; standard-library trace. Branch coverage not measured.
Dataset validator	Passed all nine CSVs, 21 knowledge documents, joins, order math, practice evidence and verification gates.

Results and remaining review

The full Step 1/2 asr_text comparison and owner human review are complete. Retrieval evidence passed 47/47 scored Step 2 cases; required-fact screening passed 29/49 after 10/49 in Step 1. No fresh full Step 3 or speech-batch API run was made for this update.

Reproduce locally

Purpose	Command
Install	`python -m pip install -r requirements.txt`
Validate data	`python validate_dataset.py`
Interactive agent	`python run.py --step 3`
Local website	`python web_app.py -> http://127.0.0.1:8000`
Audio folder	`python batch_run.py --output sample_outputs`
Full question scores	`python evaluate.py --dataset dataset --steps 1 2 --output reports/asr-full.jsonl`

Keep credentials private. Keys belong in ignored local .env. Dataset copies, API keys, virtual environments and the personal mic recording are excluded from the submission ZIP.

GitHub: https://github.com/vanshdixit1/zentromart-voiceops | Live site: https://zentromart-voiceops.vercel.app | Text agent: python run.py --step 3